Generative Models

- In supervised learning, we have data x and target output y, and the goal is to learn a function to map x to y
- While in unsupervised learning, there are no labels and the goal is to find some underlying hidden structure of the data
- A Generative Model is a powerful way of **learning any kind of data distribution using unsupervised learning**
- All types of generative models aim at learning the true data distribution of the training set so as to **generate new data points** with some variations

	Generative Model	Discriminative Model
Informally	Generate new data instances	Discriminate between different kinds of data instances
Formally	Capture the joint probability $p(X, Y)$ or just $p(X)$ if there are no labels	Capture the conditional probability $p(Y\|X)$

Generative Adversarial Network (GAN)

Training algorithm overview

- Initialize generator and discriminator
- In each training iteration:
    - Step 1: Fix generator G and update discriminator D
      The discriminator will learn to assign high scores to real objects and low scores to generated objects
    - Step 2: Fix discriminator D and generator G
      The generator learns to “fool” the discriminator, the goal is to make the discriminator believe the generated image is real

GAN as Structured Learning

- **Structured Learning:** Output multiple connected parts, such as a sequence, a matrix, a graph, a tree, … (Output is composed of component with dependencies)
- Example: Machine translation, Speech recognition, Chat-box, Image to Image translation, Text to Image translation

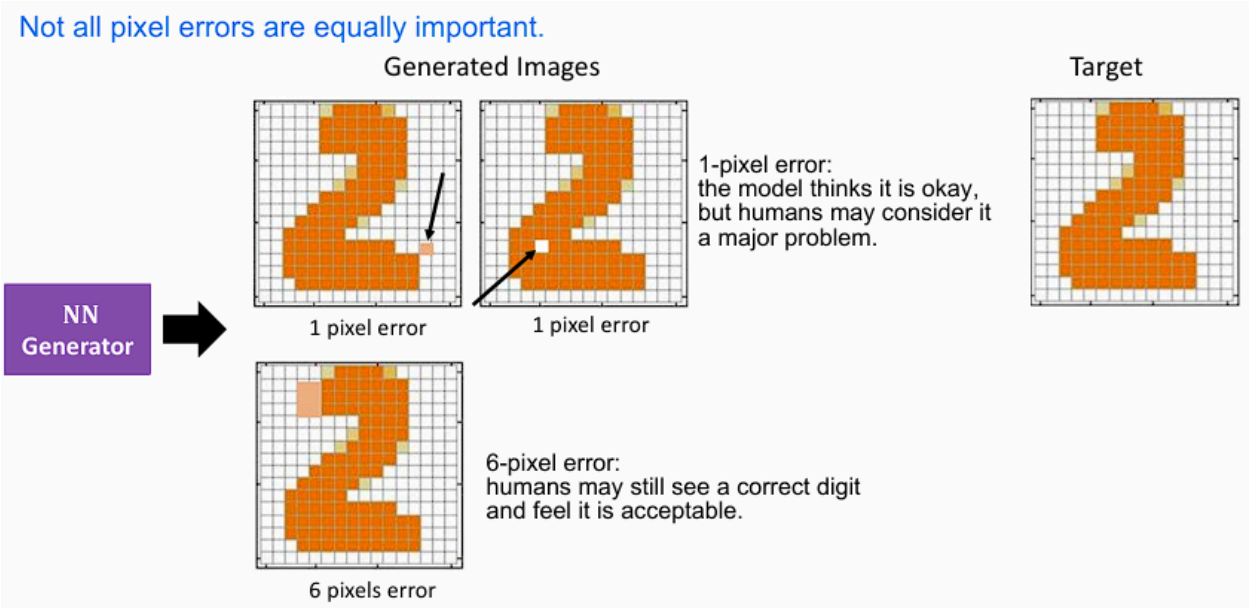
Why is structured learning challenging?

- One shot / Zero shot learning:
  - In classification, each class has some examples
  - In structure learning:
    - If we consider each possible output as a “class”
    - Since the output space is huge, most “classes” do not have any training data
    - Machine has to create new stuff during testing
    - Need more intelligence
- Machine has to learn to do planning
  - Machine generates objects component-by-component, but it should have a big picture in its mind
  - Because the output components have dependency, they should be considered globally
  - For example:
    - A pixel of image might depend on its neighboring pixels
    - The next word in a sentence is dependent on the sequence of the previous words

GAN is a combination of:

- Generator (Bottom Up): Learn to generate the object at the component level
- Discriminator (Top Down): Evaluating the whole object, and find the best one

Without a discriminator, we can use Auto-encoder (NN Encoder and NN Decoder learn together). This will be fine if the generator can truly copy the target image. However, this will not always be the case, and if the generator makes errors when generating an image:



⇒ Structured learning needs to understand relationships between parts, not just compare pixels one-by-one

- The relation between components which is critical will be missed out
- The highly correlated components are not able to influence each other
- Hence, we need a deep structure to catch the relation between components

Discriminator Training

- If we don't use a generator, the discriminator only manage to learn to output "1" (real) as we only have real images
- Given a set of positive examples, and randomly generate a set of negative examples
- In each iteration:
  - Learn a discriminator D that can discriminate positive and negative examples
  - Generate negative example by discriminator by solving $\bar{x} = argmax_{x \in X} D(x)$ which aims to maximize the score of classifying real and generated images
- However, this function turns out to be too complicated to address, if we do not define any limitation.
- Since this limitation will also restrict the model's capacity, people find it easier to replace solving the $argmaxD(x)$ function with a separate generator

	Generative model	Discriminative model
Pros	Easy to generate especially with deep model	Consider the big picture
Cons	Imitate the appearance Hard to learn the correlation between components	Generation is not always feasible How to do negative sampling?

⇒ Benefit of GAN:

- From Discriminator's point of view: Using generator to generate negative samples
- From Generator's point of view: Still generate the object component-by-component, but it is learned from the discriminator with global view

Limitation of GAN

- It may generate a random image from the domain. These is a relationship between points in the latent space to the generated images, but this relationship is complex and hard to map
- Some datasets have additional information, such as a class label, and it is desirable to make use of this information
- There are 2 motivations for making use of the class label information in a GAN model:
    - Improve the GAN
    - Targeted Image Generation

Conditional GAN

- Text-to-Image:
    - Traditional supervised learning's problem: Greater possibility to produce blurred images. (maybe due to the average of several images)
    - If the discriminator model is trained in such way:
        - Discriminator takes in 2 inputs which are the input and output of the generator
        - Then output a scalar
          (x is a real image or not + c and x are matched or not)

      ⇒ The generator will learn to generate realistic images
      but completely ignore the input conditions
- Image-to-Image:
    - Same to Text-to-Image
    - It is better to use GAN + similarity loss
- Video Generation

Unsupervised Conditional Generation

Transform an object from 1 domain to another without paired data (style transfer)

- Approach 1: Direct transformation
- Approach 2: Projection to common space